Ganga Meghanath

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Profile

Applied scientist with 6+ years across data science, machine learning, and production AI systems. Most recently lead scientist at Amazon on a multi-agent LLM system that detects authorization-bypass vulnerabilities in application code. Designed the pipeline end to end and owned its evaluation: historical and live-code benchmarks, precision measured under triage by security engineers, and the failure analysis that guided each round of coverage improvements. Before that, four years at Microsoft across Windows Experimentation and Bing Ads, spanning large-scale causal inference and A/B testing on device telemetry, and production ML for advertising policy enforcement and fraud detection.

Technical Skills

LLM & Agents	Amazon Bedrock, smolagents, Strands Agents SDK, Pydantic, Semantic Kernel, RAG, retrieval and reranking; Claude, GPT, Llama, DeepSeek-R1, Nova, Palmyra, Cohere
Evaluation	Benchmark design under sparse ground truth, held-out and out-of-time validation, sampled expert triage, ablation studies, model comparison, failure analysis
Experimentation	Experimental design, significance testing, A/B testing; causal discovery (DirectLiNGAM, DECI); causal inference (Double Machine Learning, synthetic difference-in-differences, BSTS)
Programming & ML	Python, SQL, Kusto, Pandas, NumPy, Spark; PyTorch, TensorFlow, scikit-learn, LightGBM, Hugging Face, EconML, Causica, ONNX
Platforms	AWS (Bedrock, CDK), Azure ML, Git, Linux, Power BI

Experience

Amazon Security — New York, NY

Applied Scientist II, Core Infrastructure Security — *Jan. 2026 – Jul. 2026*

- **Multi-agent vulnerability discovery, as lead Applied Scientist.** Architected a system that detects authorization-bypass vulnerabilities in application code: designed code- and application-context enrichment and risk prioritization by likelihood and impact, and integrated team-built discovery, data-flow, authorization-reasoning, and critic agents through a typed orchestration layer (Strands Agents SDK, Pydantic contracts). Originated the approach in a seven-person hackathon whose proof of concept became the official workstream.
- **Evaluation of that system.** Designed two independent tracks so results did not rest on a single metric. Held-out historical evaluation against pre-fix code recovered **13 of 24** known vulnerabilities across 17 packages, every miss traced to attack-surface coverage. Live-code evaluation across 10 applications produced 179 findings; three security engineers triaged matched samples from three of them, indicating approximately **86% precision** and an estimated **3× higher true-positive yield** than an existing agentic scanner. Failure analysis guided successive coverage improvements.
- **AWS architecture diagrams from infrastructure-as-code, a separate four-person project.** Built the resource-connectivity agent using smolagents CodeAgent, custom traversal tools, and intermediate caching. Evidence-backed evaluation on a 10-package benchmark gave approximately **99% mean per-package precision** and **93% estimated conditional recall** among upstream-detected resources; led evaluation and builder pilots, and the system reached preview integration internally.

Data Scientist II, Proactive Security Organization — *May 2025 – Dec. 2025*

- **Security finding classification (RAG).** Designed and experimentally validated the production architecture for a retrieval-augmented pipeline mapping security findings across roughly **940 weakness categories** on Amazon Bedrock, pairing Cohere embedding retrieval with Claude classification and a critique pass: approximately **80% precision**, **95% top-10 retrieval accuracy**.
- Selected it through model comparisons and ablation studies against Llama, Palmyra, Nova, DeepSeek-R1 and sentence-transformer baselines, and dropped reranking from the production design once it stopped earning its runtime cost; handed off a CLI adopted by a central security-finding platform.
- **Evaluation practice across the security org.** Used the project as a worked example for engineers building their own LLM systems: setting a baseline and improving on it, designing ablations, reading configuration effects, and why held-out evaluation sets and unseen production data give different answers.

Microsoft, Windows Experimentation — Redmond, WA

Data Scientist I → II (Promoted June 2024) — *Feb. 2023 – Apr. 2025*

- **Windows 11 rollout safety.** Led telemetry analysis for the 2024 rollout, combining pattern mining, LightGBM, and causal discovery over noisy device telemetry to identify configurations linked to rollbacks and crashes, including previously unreported causes that engineers reproduced and fixed. Produced deployment blocklists covering **5.6M early-adopter** and approximately **40M general-retail devices**.

- **Internal RAG assistant.** Built a retrieval-augmented assistant answering experimentation questions over platform documentation, using GPT-Turbo with Ada embeddings, Semantic Kernel, and Azure Cognitive Search; delivered the proof of concept and its production plan.
- **Causal analysis across Windows teams.** Subject-matter resource for the platform, advising six internal and customer projects on study design, causal discovery (DirectLiNGAM, DECI), and causal inference (Double Machine Learning, synthetic difference-in-differences, BSTS); built an Azure ML DML pipeline with automated feature transformation and imputation, transferred for productionization.

Microsoft, Windows Data
Redmond, WA

Data Scientist Intern — *May 2022 – Aug. 2022*

- **Personalized Windows 11 Settings recommendations.** Built a pointwise LightGBM ranker under on-device model constraints, improving coverage from **65% to 75%** on a six-month out-of-time test set; the prototype later shipped through a controlled feature rollout.

Purdue University, Makin Lab
West Lafayette, IN

Graduate Research Assistant — *May 2021 – Dec. 2022*

- **Neural population dynamics.** Developed RNN, Transformer, and TERN (Transformer Encoder over RNN) models to infer population dynamics on Neural Latents Benchmark datasets; ranked **first on 2 of 4 benchmark datasets** and led the resulting first-author *Journal of Neural Engineering* publication.
- **Decoding speech from brain activity.** Reconstructed spoken sentences from electrocorticography (ECoG) recordings in patients with surgically implanted electrode arrays, using sequence-to-sequence GRUs with cross-attention and pretrained Transformer decoders; matched prior performance of approximately **3% word error rate** for one subject at word-level segmentation, and tested generalization to unseen sentences under limited data.

Microsoft, Bing Ads
Bengaluru, India

Data & Applied Scientist — *Jun. 2019 – Dec. 2020*

- **Advertising policy-violation detection.** Developed and deployed classification pipelines using **MobileNet, ResNet, and EfficientNet** image classifiers with ONNX/C# inference tooling, including a production classifier that improved precision from **5% to 40%** and recall from approximately **30% to 80%**.
- **Fraud-risk feature store.** Built a **367-feature** store refreshed every 15 minutes with a 10–15 minute SLA for Spark and Cosmos classifiers, decoupling shared feature computation from the downstream detection models; extending account-reputation signals across 28 algorithms cut notification volume by approximately **13%** in offline analysis.
- **Account-compromise detection.** Modelled advertiser-account takeover from newly added domains, scoring each against the account's domain history using domain embeddings, reputation, and search-index signals, with separate treatment for changes crossing top-level domains; deployed the production feature lookup store serving those signals.

Publications

Ganga Meghanath, Bryan Jimenez, and Joseph G. Makin. "Inferring population dynamics in macaque cortex." *Journal of Neural Engineering*, 20(5):056041, 2023. **First author.**

Ran Xu, Rakesh Kumar, Pengcheng Wang, Peter Bai, **Ganga Meghanath**, et al. "ApproxNet: Content and contention-aware video object classification system for embedded clients." *ACM Transactions on Sensor Networks*, 18(1), 2021.

Public Work, Talks and Teaching

Diffusion Models in PyTorch: Modular implementation for MNIST, Fashion-MNIST, and CIFAR-10, with experiments on diffusion length, variance schedules, and loss functions, evaluated by FID and Inception Score.

Causal Analysis Tutorial: Open-source Python tutorial on causal discovery and Double Machine Learning with Causica, LiNGAM, and EconML, designed and delivered as a hands-on workshop at Women in Data Science, Seattle University, 2024.

Causal Analysis Overview and Beginner's Guide to Causal Discovery: Technical articles in Microsoft's Data Science publication on choosing between experimentation, causal inference, and causal discovery.

Agent 403: Access Denied, internal security research conference, 2026. Poster creator and primary presenter.

Technical community at Microsoft. Co-organized DS3, a monthly Data Science & Statistics seminar for the Windows organization, around eight sessions a year with internal and external speakers; co-presented on causal discovery and inference at MLADS 2024, with a companion poster, and gave a WinLearn talk on causal analysis.

Education

Purdue University — West Lafayette, IN

M.S., Electrical and Computer Engineering; GPA: 4.0/4.0 — *Jan. 2021 – Dec. 2022*

Indian Institute of Technology Madras — Chennai, India

B.Tech., Electrical Engineering; Minor in Data Science; CGPA: 8.79/10.0 — *Aug. 2015 – May 2019*